

Ten-Layer Controlled-Viability Review

RH-130–RH-139 and a Revised Minimal Directional Frontier

Prime Dynamics Theory Program

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Abstract

We review RH-130–RH-139 as one proof system. The ten layers remove the positive-Gram floor, treat singular support, derive a model-generated moving frame, expose target-metric amplification, solve the orthogonal metric minimax problem, replace stationary fixed-floor optimization by a finite-horizon control envelope, and close the frozen finite chain with outward Loewner guards. The finite progression is substantial: the initial floor-free transfer has zero positive complete chains, the Euclidean polar gauge has only 51/216 contractive recurrent transitions, metric balancing recovers 183/216, finite-horizon control certifies 328/330 transitions and 28/30 chains, and outward composition retains all 28 positive chains.

The main theoretical conclusion is a weaker and more accurate all-level frontier. Let y_n be any validated controlled upper for the squared relative tail and let $a_n = \sqrt{\lambda_{\min}(G_n)/\lambda_{\max}(G_n)}$. If

$$\limsup_{n \rightarrow \infty} y_n \leq \bar{y} < 1, \quad \liminf_{n \rightarrow \infty} a_n \geq a_* > 0,$$

then the directional candidates obey

$$\liminf_{n \rightarrow \infty} B_n \geq (1 - \sqrt{\bar{y}})^4 a_* > 0.$$

This theorem does not require a subunit affine coefficient at every step; isolated expanding transitions may be crossed by a state-dependent viability policy. The two hypotheses are separately sharp within the directional candidate architecture: allowing $y_n \rightarrow 1$ or $a_n \rightarrow 0$ destroys every positive universal floor.

Thus the exact-matrix mathematical frontier is reduced to two packets: eventual uniform controlled-tail viability and a positive normalized-base liminf. A validated model route additionally needs exact or interval source enclosure; independently assembled routes also need all-level outward radii. All nine upstream archives and claim boundaries pass an automated review. None of the three missing all-level packets is proved here.

1 What changed over ten layers

RH-129 identified an all-level subunit affine Rayleigh recurrence and a positive normalized-base liminf as the directional frontier. RH-130–RH-138 reopened the first packet from the

physical packet construction rather than from post-hoc regularized Gramians. Three changes proved decisive.

First, singularity is structural rather than numerical. RH-130 and RH-131 show that the recent action may have zero tail rank followed by rank birth, so the correct quotient lives on the supported subspace. Second, the raw memory coefficient is strongly contractive but the recent-Gram metric is not: RH-134 gives a raw factor below 0.003914, whereas RH-135 finds a polar-gauge relative base as large as 3.03×10^7 . Third, long-run contractivity is not the same as finite viability. RH-136 proves 33 transitions are noncontractive for every orthogonal gauge, but RH-137 crosses 31 of them safely because their incoming certified state is small.

This last observation changes the proof obligation. The route does not need $\rho_n < 1$ at every level. It needs a controlled state that eventually stays a uniform distance below the support wall. This is a viability condition in the standard control-theoretic sense [Aubin, 1991], specialized here to a scalar order-preserving envelope.

2 Controlled viability replaces per-step contraction

At level n , let $x_n = \gamma_n^2$ be the actual squared relative tail. A finite family of packet gauges gives monotone Young envelopes

$$F_{n,c}(y) = q_{n,c} + \left(\sqrt{A_{n,c}y} + \sqrt{B_{n,c}} \right)^2, \quad c \in \mathcal{C}_n.$$

If $y_n \geq x_n$, define the controlled upper

$$y_{n+1} = \inf_{c \in \mathcal{C}_n} F_{n,c}(y_n). \quad (1)$$

RH-137 proves that for a finite candidate family this greedy choice minimizes every later finite-horizon envelope. RH-138 shows how two outward residuals can validate each chosen transition from rounded matrices and radii.

Let

$$a_n = \frac{\sqrt{\det G_n}}{L_n^4 C_n} = \sqrt{\frac{\lambda_{\min}(G_n)}{\lambda_{\max}(G_n)}}$$

be the normalized directional base. The candidate lower is

$$B_n = (1 - \sqrt{x_n})_+^4 a_n. \quad (2)$$

Theorem 1 (Eventual controlled-viability support). *Suppose $x_n \leq y_n$ eventually and*

$$\limsup_{n \rightarrow \infty} y_n \leq \bar{y} < 1, \quad \liminf_{n \rightarrow \infty} a_n \geq a_* > 0.$$

Then

$$\liminf_{n \rightarrow \infty} B_n \geq (1 - \sqrt{\bar{y}})^4 a_* > 0.$$

Consequently, every fixed threshold strictly below this floor is eventually supported by the directional candidate.

Proof. Since $x_n \leq y_n$ and $z \mapsto (1 - \sqrt{z})_+^4$ is decreasing,

$$B_n \geq (1 - \sqrt{y_n})_+^4 a_n.$$

For every $\varepsilon > 0$, eventually $y_n \leq \bar{y} + \varepsilon < 1$ and $a_n \geq a_* - \varepsilon$. Taking the lower limit and then $\varepsilon \downarrow 0$ proves the formula. $\square$

The theorem strictly weakens the RH-128 route. A contractive affine recurrence is one sufficient method for constructing the required tube, but it is not part of the conclusion and need not hold at isolated steps. For example, $F(y) = My$ has coefficient $M > 1$ yet crosses safely whenever $y < 1/M$. What matters is the global controlled trajectory, not the sign of one local logarithmic multiplier.

Proposition 2 (Architecture-sharp frontier). *Within the candidate formula (2), neither the strict uniform tail gap nor the positive base liminf can be removed from a universal positive eventual lower.*

Proof. For the tail-gap obstruction, take $y_n = x_n = (1 - 1/n)^2$ and $a_n = 1$. Then $y_n \uparrow 1$ and $B_n = n^{-4} \rightarrow 0$. For the base obstruction, take $x_n = y_n = 0$ and $a_n = 1/n$. Again $B_n \rightarrow 0$. Both examples satisfy the other packet in its strongest form. $\square$

This is an architecture-relative necessity statement, not a claim that every possible proof of a broader theorem must use these variables.

3 Layer-by-layer audit

Table 1 records the logical role of each paper. “Mixed” means that a constructive theorem and a sharp obstruction are equally central.

The automated review reads every upstream summary and archive-verification record, hashes the nine summaries, and inspects the explicit program boundaries. All nine archives verify and no Stage A, Hilbert–Polya, or RH flag is asserted. A 4,096-instance scalar audit of Theorem 1 has zero floor failures. The two sharp witness families are also archived.

4 What the finite result now says

The frozen packet family has moved from no complete floor-free chains to 28 outward-positive chains. This improvement is not produced by one optimistic regularization. It comes from four identifiable corrections: supported-subspace quotients, the exact memory recursion, metric-aware gauge selection, and state-dependent finite-horizon optimization. RH-138 then shows that 40-decimal independent rounding and explicit norm radii preserve the result with negligible outward padding.

The terminal directional lowers of the 28 positive chains range above 10^{-10} ; 21 exceed 10^{-8} , 16 exceed 10^{-6} , and 12 exceed 10^{-4} . The two failed chains are coarse left-channel cases whose new boundary term is already $9.40589 > 1$ relative to the target Gram. They do not show that an eventual-start theorem is impossible: an asymptotic route may begin

RH	type	exact contribution	finite consequence or obstruction
130	mixed	floor-free semidefinite minimax	118/120 positive states, but 0/24 complete chains; 24 rank births
131	constructive	support quotient and pseudovolume	kernel leakage exactly characterizes infinite relative quotients
132	constructive	canonical partial-isometry forcing	trace-minimal forcing; 22/24 birth strengths subunit
133	mixed	dyadic packet gauge	65 versus 67 positive transfers; angle-only bounds fail sharply
134	constructive	exact memory/birth recurrence	330/330 identities and Loewner bounds; birth dominates forcing
135	mixed	sharp relative affine coefficients	polar gauge contractive on only 51/216 recurrent steps
136	constructive	orthogonal metric minimax	183/216 possible and recovered; 33 universal orthogonal walls
137	constructive	pointwise Young viability control	31/33 walls crossed; 328/330 safe steps and 28/30 chains
138	constructive	outward residual/base composition	330/330 residual pairs; 28 outward-positive chains; precision gate
139	synthesis	controlled-viability frontier	per-step contraction removed; two sharp remaining mathematical packets

Table 1: The RH-130–RH-139 proof stack.

after finitely many coarse events. They do show that a certificate starting from the present rank-four/depth-eight initialization cannot retain all 30 chains.

The finite precision result has a similarly exact boundary. Twenty decimal digits suffice to keep every archived normalized base positive under a norm ball, whereas fp64 loses ten weak coarse directions. By the RH-138 norm-ball barrier, these ten losses cannot be repaired by a cleverer eigenvalue inequality without adding information [Bhatia, 1997, Horn and Johnson, 1991, Moore, 1966].

5 Revised minimal frontier

For exact all-level matrices, Theorem 1 and Proposition 2 leave the inclusion-minimal frontier

$$\boxed{P_V : \limsup y_n < 1}, \quad \boxed{P_B : \liminf a_n > 0}.$$

Here P_V may be obtained by block control, resets, delayed start, occasional metric expansion followed by strong contraction, or another monotone policy. It is strictly weaker than requiring every affine base or every optimized ρ_n to be subunit.

To connect these exact-matrix packets to the intended model, add

$$\boxed{P_E : \text{exact or interval source-to-packet enclosure}}.$$

If source, tail, forcing, and target matrices are assembled independently, the RH-127/RH-138 logic additionally requires

$$\boxed{P_O : \text{all-level outward radii and residual guards}}.$$

Thus the two validated routes are

$$\{P_V, P_B, P_E\} \quad (\text{common assembly}), \quad \{P_V, P_B, P_E, P_O\} \quad (\text{independent assembly}).$$

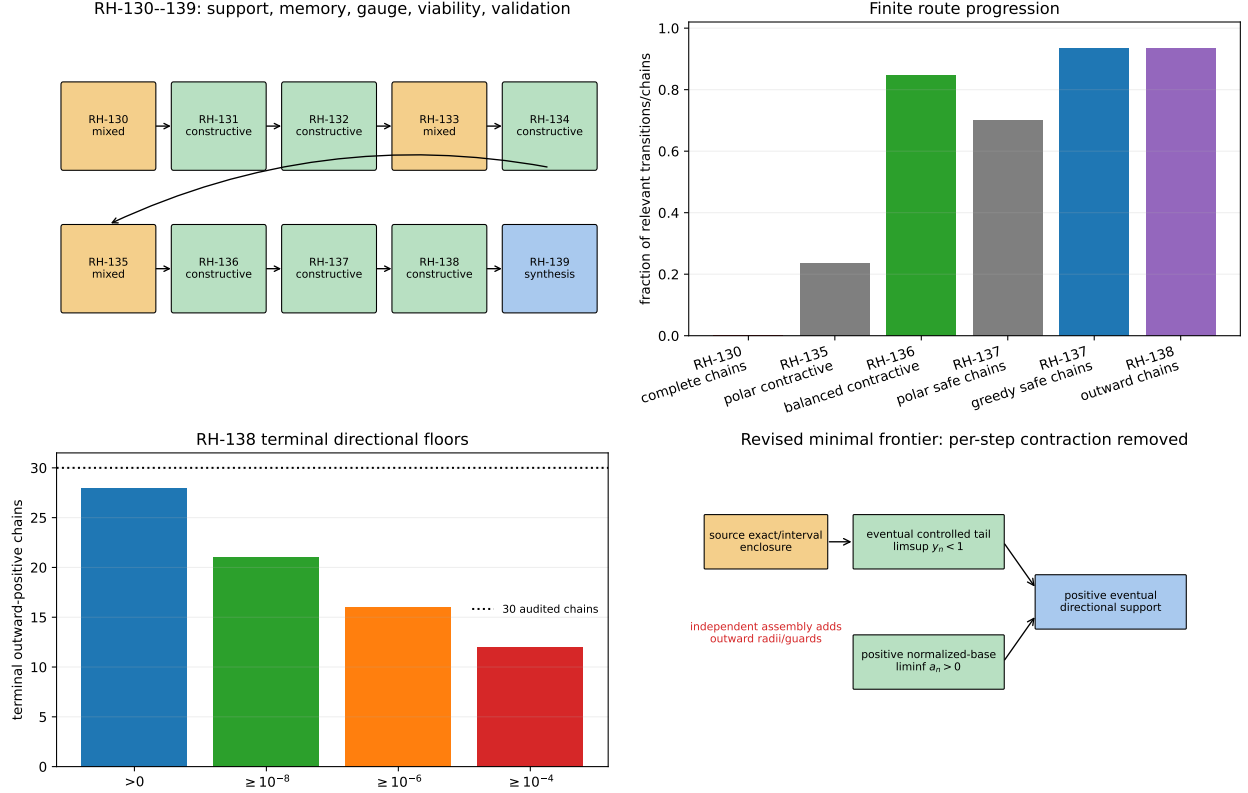


Figure 1: The ten-layer dependency path, finite progression, terminal directional floors, and the revised minimal frontier.

The current work supplies finite evidence and exact algebra for every box, but proves none of P_V, P_B, P_E all levels.

6 Next research block and claim boundary

The next block should be narrower than the preceding maze exploration. First, intervalize the source snapshots and packet/frame construction so that RH-138 validates the intended model rather than only a frozen reference. Second, search for an all-level block or reset theorem for the controlled map (1); the correct target is a uniform viability tube, not per-step contraction. Third, attack the weakest-direction base a_n on the same correlated assembly path, avoiding the separate volume/capacity loss identified in RH-124. A delayed-start analysis should determine whether the two coarse birth failures are irrelevant to an eventual statement.

Only after these packets close would the directional Stage A branch have an all-level foundation. RH-139 does not prove the controlled tail gap, the positive base $\liminf$, source-model interval enclosure, all-level outward radii, uniform Stage A, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

References

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